Esercitazioni Ing.Sw

Gian Enrico Conti

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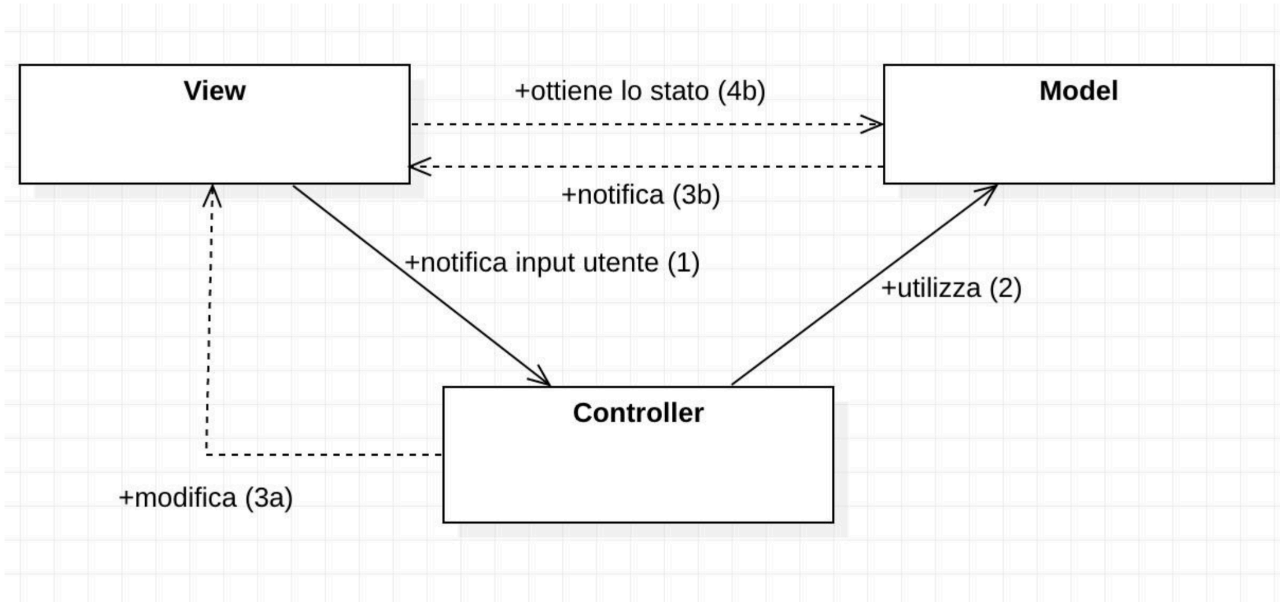
Towards Remote View

https://github.com/ingconti/TowardsRemoteView.git

LOGIC:

- MVC con M V C in locale
- I button/KBD modificano il modello
    - View -- (evento) --> controller **modifica** modello

(invocazione di metodo)



- Aggiungiamo network...

LOGIC: Aggiungiamo network

- Visto in "MinimalTCPServerAndClient"

Client manda Comandi
Server che risponde ai comandi
Server con un thread per ogni client (multi thread)
Parsing dei comandi
Mini Modello
Modello che "evolve"
Modello che "risponde"
Rete che rimanda esito al client
Client che aggiorna la vista (per ora terminale...)

Quindi....

Client->Rete->Server->(Parsing)->Modello (evolve...) -> Rete -> View (per ora cmd line.. ***sout***)

Prof: "modello (via strato di rete) va direttamente alla view/client, senza ripassare dal controller."

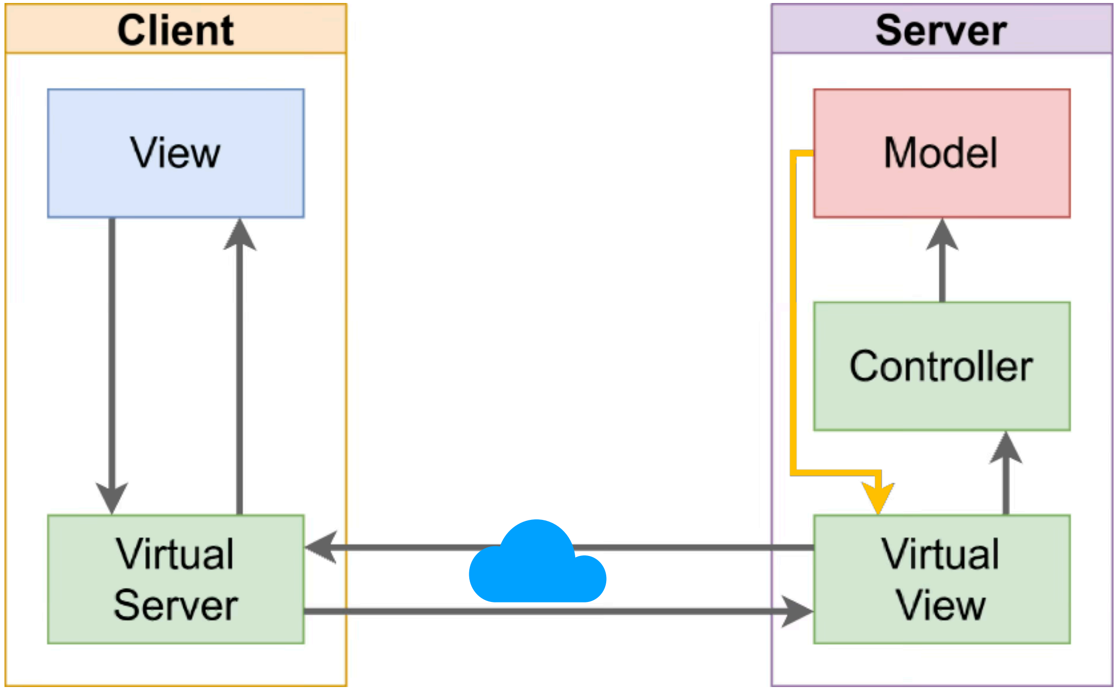
LOGIC (simpler)

- Modello su server!
- Network
- Invocazione di metodo sul modello
- Ricevo risposta
- Mando via network
- Sul client ricevo -> View

Full schema (simpler)

Dal modello generano RISPOSTE e vanno lungo la rete:

Connessione "yellow"



(i.e. "Client->Rete->Server->(Parsing)->Modello (evolve...) -> Rete -> View (per ora cmd line.. ***sout***)"

LISTENERS

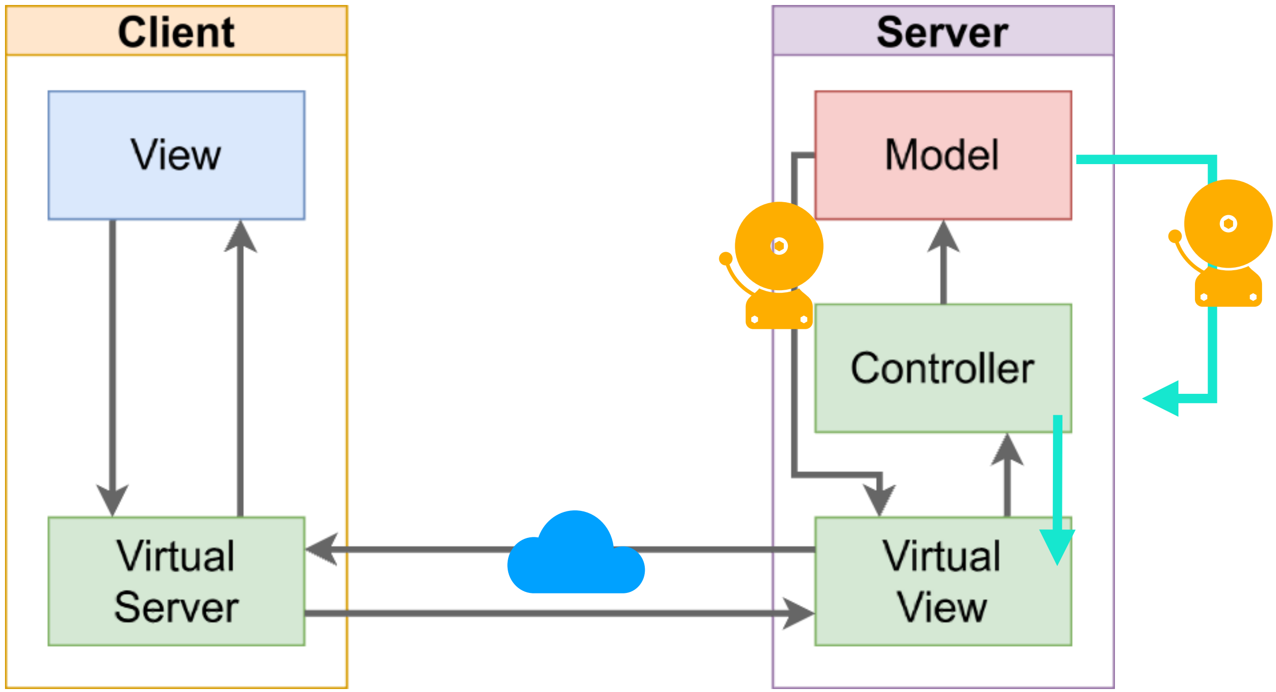
- Modello su server!
- Client manda
- Listeners ANCORA su modello (i.e. sul server!)
- Listener propagano -> View (virtual) -> rete -> update (sul client)

Dal modello generano RISPOSTE e vanno lungo la rete:

Connessione "verde o "nera"?

Piu semplice "nera":

Listener su VV.



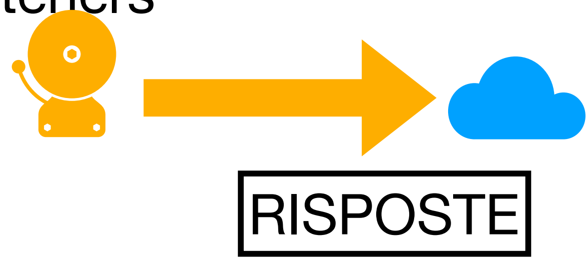
STEPS:

- app (client) in JavaFX ("TowardsRemoteViewDemo")
- Modello ("*Restaurant*" *FSM*)

Testiamo in locale

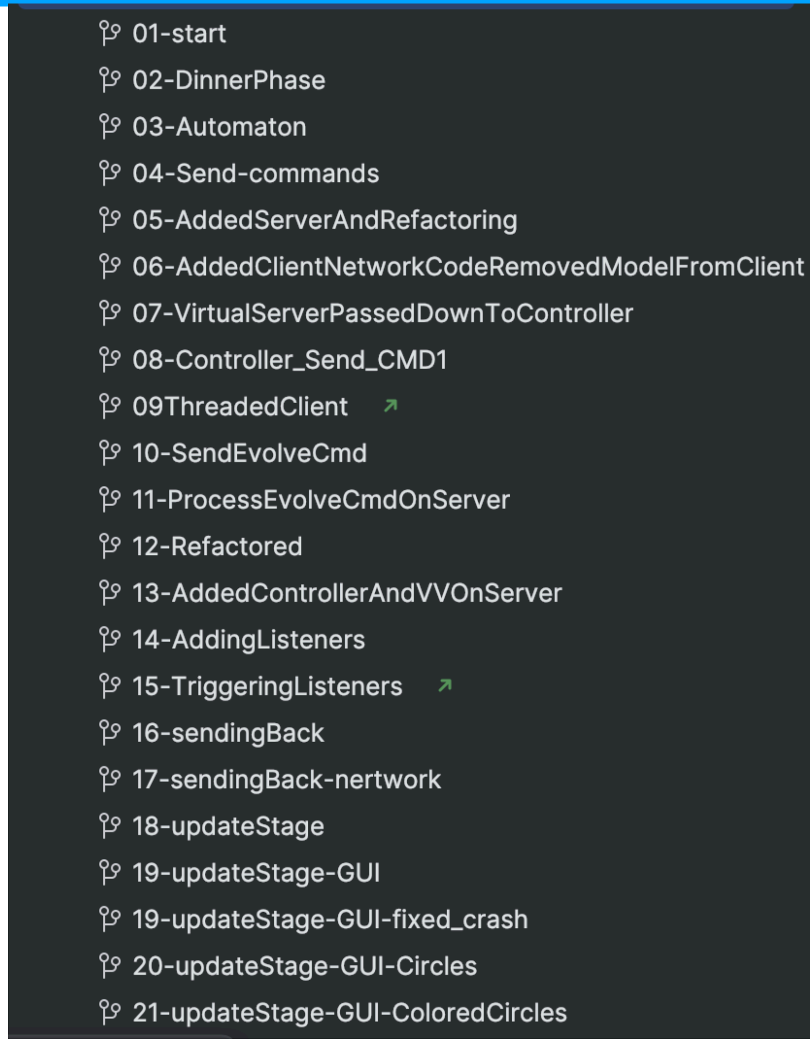
..

- Miniserver
- Testiamo "m ver" co saggi testuali da terminale
- Spostiamo mo er
- Codifichiamo i c nessaggi
- Messaggi --- -> nel modello -> Listeners



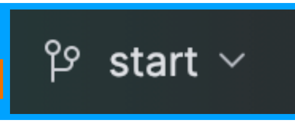
- For your convenience:

(TowardsRemoteViewDemo)



app (client) in JavaFX: Modify Controller

1



add an attribute and setter:

...

```java
private Stage myStage;
public void setStage(Stage stage) {
    myStage = stage;
}
```

And fix. POM..

Some fixes..


```
public void start(Stage stage) throws IOException {
    //Was: FXMLLoader fxmlLoader = new FXMLLoader(HelloApplication.class.getResource("hello-
view.fxml"));
    FXMLLoader fxmlLoader = new FXMLLoader(HelloApplication.class.getResource("hello-view.fxml"));

    //Added:
    Parent root = fxmlLoader.load();
      HelloController controller = fxmlLoader.getController();
      controller.setStage(stage);

    //was: Scene scene = new Scene(fxmlLoader.load(), 320, 240);
    Scene scene = new Scene(root, 320, 240);

    stage.setTitle("Hello!");
    stage.setScene(scene);
    stage.show();
}
```

Il controller vede il suo stage, i.e. la VIEW

Modello

Abbiamo un semplice ristorante:

(specs &&& code for a simple network automaton here: https://github.com/ingconti/AutomatonFromNetwork)

```
Our FSM describes a lunch.
It starts from ENTREE and evolves sending "g" (GO!) command from client
on TCP.
```

States are:

```
    UNKNOWN,
    ENTREE,
    MAIN_COURSE,
    SECOND_COURSE,
    DESSERT,
    THE_END_OF_LUNCH;
```

Modello

You can try using Tests!

Allowed commands are:

- "G" (GO!)
- every initial char of every state, i.e. "E", "M"..."T"
- "P" to pay.

You cannot go back when specifying state.

NOTE: you cannot finish lunch without PAYING!
You can pay in any state you are in.

User cannot see why he cannot go on.

Note: for simplicity these commands are equal to states, (except "G").
to be precise we should pay more attention: cmd should be:
"GoTo_E", "GoTo_M" and so on...

Try it!


```
To test against console:

(Dowload form git... run.. it will listen on port 1234)


/usr/bin/nc 127.0.0.1 1234

and type in console:
server will receive and send back some info.


Live..
```

```
g
 new state: MAIN COURSE
g
 new state: SECOND COURSE
g
 new state: DESSERT
p
 new state: DESSERT
p
 new state: DESSERT
asd
 NOT MOVED FROM DESSERT
p
 NOT MOVED FROM DESSERT
p
 NOT MOVED FROM DESSERT
d
 NOT MOVED FROM DESSERT
m
 NOT MOVED FROM DESSERT
g
 new state: END OF YOUR LUNCH!
```

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Modello: ENUM "DinnerPhase"

2 DinnerPhase

```java
enum DinnerPhase implements Comparable<DinnerPhase> {
    UNKNOWN,
    ENTREE,
    MAIN_COURSE,
    SECOND_COURSE,
    DESSERT,
    THE_END_OF_LUNCH;

    DinnerPhase next() {

        switch (this) {
            case ENTREE:
                return MAIN_COURSE;

            case MAIN_COURSE:
                return SECOND_COURSE;

            case SECOND_COURSE:
                return DESSERT;

            case DESSERT:
                return THE_END_OF_LUNCH;

            case THE_END_OF_LUNCH:
                return THE_END_OF_LUNCH;
        }

        return UNKNOWN;
    }

    static DinnerPhase fromString(String s) {

        switch (s.toUpperCase().charAt(0)) {
            case 'E':
                return ENTREE;
            case 'M':
                return MAIN_COURSE;
            case 'S':
                return SECOND_COURSE;
            case 'D':
                return DESSERT;
            case 'T':
                return THE_END_OF_LUNCH;
            case 'U':
                return UNKNOWN;
        }
        return UNKNOWN;
    }

    @Override
    public String toString() {
        switch (this) {
            case ENTREE:
                return "ENTREE";
            case MAIN_COURSE:
                return "MAIN COURSE";
            case SECOND_COURSE:
                return "SECOND COURSE";
            case DESSERT:
                return "DESSERT";
            case THE_END_OF_LUNCH:
                return "END OF YOUR LUNCH!";
        }
        return "UNKNOWN";
    }
}
```

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App: add automaton


```java
public void start(Stage stage) throws IOException {

    FXMLLoader fxmlLoader = new
FXMLLoader(HelloApplication.class.getResource("hello-
view.fxml"));

    // Added:
    Parent root = fxmlLoader.load();
    HelloController controller = fxmlLoader.getController();
    //was: controller.setStage(stage, this.automaton);
    controller.setStage(stage, this.automaton);

    Scene scene = new Scene(root, 320, 240);

    stage.setTitle("Hello!");
    stage.setScene(scene);
    stage.show();
}
```

```java
// add model.
Automaton automaton;
public HelloApplication() {
    this.automaton = new Automaton();
}
```

Modello: Class "Automaton"


```java
public class Automaton {

    private Boolean paid = false;
    private DinnerPhase state = DinnerPhase.ENTREE;

    public DinnerPhase getState(){
        return state;
    }

    private Boolean canEvolve(){
        int currOrd = state.ordinal();
        int dessertOrd = DinnerPhase.DESSERT.ordinal();

        if (currOrd >= dessertOrd && !paid) {
            System.out.println("PAY BEFORE!!!");
            return false;
        }
        return true;
    }

    public void setPaid(){
        paid = true;
    }

    public Boolean evolve() {
     if (!canEvolve())
            return false;

        int currOrd = state.ordinal();
        int lastOrd = DinnerPhase.THE_END_OF_LUNCH.ordinal();

        if (currOrd < lastOrd) {
            state = state.next();
            return true;
        }
        return false;
    }


    public Boolean evolveTo(DinnerPhase toState){
        if (!canEvolve())
            return false;

        int toOrd = toState.ordinal();
        int currOrd = state.ordinal();

        if (toOrd>currOrd) {
            state = toState;
            return true;
        }
        return false;
    }

}
```

Send commands

Come detto, dal controller chiamata di metodo al modello, che è visto per ref:

*(Nota: il button è **già** nella view (stage..) e manda touch al controller)*

```
protected void onHelloButtonClick() {
   //was: welcomeText.setText("Welcome to JavaFX Application!");
    this.automaton.evolve();
    this.updateView();
}

void updateView(){
    String status = this.automaton.getState().toString();
    this.welcomeText.setText(status);
}
```

```
Run...
```

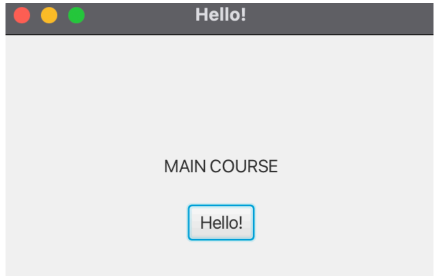
Send commands: Run

4 Send-commands

Potremmo fare un po' di refactor / renaming.. ai pulsanti...

aggiungere check su **canEvolve**,

Dare piu diagnostica..



Network!

```
/Users/ingconti/Library/Java/JavaVirtua
PAY BEFORE!!!
PAY BEFORE!!!
```

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Network

Aggiungere network al client

Creare TCP server

Test scambio messaggi

Spostare model sul server

Network base code (server)

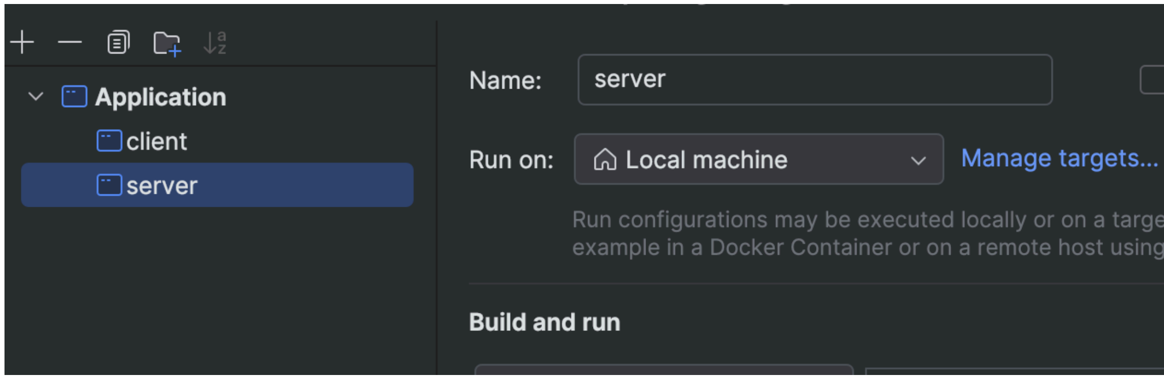
5 AddedServerAndRefactoring

(See: MinimalTCPServerAndClient2025 Slides / GitHub MinimalTCPServerAndClient 02-client-server_fixed *)*

- nuova classe ServerMain (copiata da MinimalTCPServerAndClient)

X ora NON threaded

- (rifattorizziamo HelloApplication in ClientMain)
- (rifattorizziamo HelloControlller in Controller)
- creiamo 2 config:



Network base code (client)

(See: MinimalTCPServerAndClient2025)

- Rimuoviamo model (andrà su server)
- Nuova classe VirtualServer
- Metodo "start()"
- Copiamo intero corpo di **main** di "ClientMain" / incolla dentro "start"

..

Network base code (client cont'd)

```java
public class VirtualServer {

    void start() {
        String hostName = "127.0.0.1";
        int portNumber = 1234;

        Socket echoSocket = null;
        try {
            echoSocket = new Socket(hostName, portNumber);
        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }

        PrintWriter out = null;
        BufferedReader in = null;
        BufferedReader stdIn = null;
        try {
            out = new PrintWriter(echoSocket.getOutputStream(), true);
            in = new BufferedReader(
                    new InputStreamReader(echoSocket.getInputStream()));

            stdIn = new BufferedReader(
                    new InputStreamReader(System.in));

        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }

        String userInput = "";
        while (true) {
            try {
                if (!((userInput = stdIn.readLine()) != null)) break;
                out.println(userInput);
                System.out.println("echo: " + in.readLine());
            } catch (IOException e) {
                throw new RuntimeException(e);
            }
        } // while
    }
}
```

..

Network base code (client cont'd)

- Instanziamo VirtualServer dentro client:

```
public class ClientMain extends Application {

    @Override
    public void start(Stage stage) throws IOException {
        FXMLLoader fxmlLoader = new
                FXMLLoader(ClientMain.class.getResource("hello-view.fxml"));

        Parent root = fxmlLoader.load();
        Controlller controller = fxmlLoader.getController();
        // removed controller.setStage(stage, this.automaton);
       controller.setStage(stage);
       Scene scene = new Scene(root, 320, 240);
       stage.setTitle("Hello!");
       stage.setScene(scene);
       stage.show();

       // added:
       this.virtualServer.start()
    }

    public static void main(String[] args) {
        launch();
    }


    /*removed
     Automaton automaton;
    public ClientMain() {
        //removed this.automaton = new Automaton();
    }
     */

     VirtualServer virtualServer;
    public ClientMain() {
        this.virtualServer = new VirtualServer();
    }
}
```

Lanciamo QUI

Network base code (client cont'd)

AddedClientNetworkCodeRemovedModelFromClient

6

- Rimuoviamo istanza Model da Controller

```
public class Controlller {
    @FXML
    private Label welcomeText;

    @FXML
    protected void onHelloButtonClick() {
       /*removed:
        welcomeText.setText("Welcome to JavaFX Application!");
        this.automaton.evolve();
        */
        this.updateView();
    }

    void updateView(){
        /* removed
        String status = this.automaton.getState().toString();
        this.welcomeText.setText(status);
        */
    }

    private Stage stage;
    //was: public void setStage(Stage stage, Automaton automaton) {
        public void setStage(Stage stage) {
        stage = stage;
        //removed: this.automaton = automaton;
    }
    //removed: private Automaton automaton;
}
```

RUN...

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Network Client RUN

AddedClientNetworkCodeRemovedModelFromClient

6

- Crash!

```
/Users/ingconti/Library/Java/JavaVirtualMachines/openjdk-
java.net.ConnectException: Connection refused 127.0.0.1

Process finished with exit code 1
```

Prima va lanciato server...

Run again.. Good.

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Network Client: sending commands..

VirtualServerPassedDownToController

7

Passiamo virtualServer al controller:

```
public void start(Stage stage) throws IOException {
    FXMLLoader fxmlLoader = new
            FXMLLoader(ClientMain.class.getResource("hello-view.fxml"));

    Parent root = fxmlLoader.load();
    Controlller controller = fxmlLoader.getController();
    //was: controller.setStage(stage);
     controller.setStage(stage, this.virtualServer);

    Scene scene = new Scene(root, 320, 240);
    stage.setTitle("Hello!");
    stage.setScene(scene);
    stage.show();

    // added:
    this.virtualServer.start();
}
```

Cosi dal pulsante possiamo invocare metodi su VV ...

```
protected void onHelloButtonClick() {
    String msg = new Date().toString();
    this.virtualServer.sendCmd(msg);
    this.updateView();
}
```

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Network Client: send 1st command

Controller_Send_CMD1

8

All code for VirtualServer:

```
public class VirtualServer {

    PrintWriter out = null;
    BufferedReader in = null;

    void start() {

        String hostName = "127.0.0.1";
        int portNumber = 1234;

        Socket echoSocket = null;
        try {
            echoSocket = new Socket(hostName, portNumber);
        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }
        /* moved up, class instances:
        PrintWriter out = null;
        BufferedReader in = null;*=
        BufferedReader stdIn = null;
        try {
            out = new PrintWriter(echoSocket.getOutputStream(), true);
            in = new BufferedReader(
                    new InputStreamReader(echoSocket.getInputStream()));

            stdIn = new BufferedReader(
                    new InputStreamReader(System.in));

        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }
        /* removed:
        String userInput = "";
        while (true) {
            try {
                if (!((userInput = stdIn.readLine()) != null)) break;
                out.println(userInput);
                System.out.println("echo: " + in.readLine());
            } catch (IOException e) {
                throw new RuntimeException(e);
            }
        } // while*/
    }

    void sendCmd(String cmd){
        out.println(cmd);
    }
}
```

All code Controlller:

```
public class Controlller {
    @FXML
    private Label welcomeText;

    @FXML
    protected void onHelloButtonClick() {
        String msg = new Date().toString();
        this.virtualServer.sendCmd(msg);
        this.updateView();
    }

    void updateView(){

    }

    VirtualServer virtualServer;
    private Stage stage;
        public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;
    }

}
```

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Network Client: send 1st command: RUN

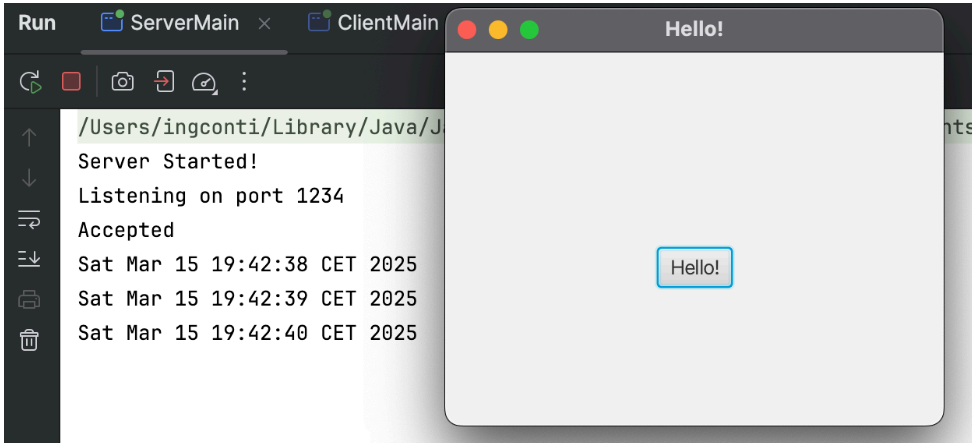
Controller_Send_CMD1

8

Run server... run client..

Click multiple times and see server..

(We do not process yet answer from server to client..)



Process answer:

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Network Client: manage reading

Nel client NON possiamo semplicemente riabilitare loop:

```
String userInput = "";
while (true) {
    try {
        if (!((userInput = stdIn.readLine()) != null)) break;
        out.println(userInput);
        System.out.println("echo: " + in.readLine());
    } catch (IOException e) {
        throw new RuntimeException(e);
    }
} // while
```

Perchè bloccante *(provateci..)*

Thread..

Creiamo classe ClientThread:

```
public class ClientThread extends Thread {
    private BufferedReader reader;

    public ClientThread(BufferedReader reader) {
        this.reader = reader;
    }

    public void run() {
        System.out.println("ClientThread started");
       int i = 0;

        while (true) {
            try {
                String answer = this.reader.readLine();
                System.out.println(answer);
            } catch (IOException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Passiamo reader,
rifastorrizzare con Socket?

…

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Network Client: manage reading: Thread (client cont'd)

09ThreadedClient

Nel VV:

...

```
  ClientThread clientThread = new ClientThread(in);
    clientThread.start();
} // end of start

void sendCmd(String cmd){
    out.println(cmd);
}
```

Run...

```
/Users/ingconti/Library/Java/
ClientThread started
SUN MAR 16 07:56:15 CET 2025
SUN MAR 16 07:56:16 CET 2025
```

Full code:

```
public class VirtualServer {

    PrintWriter out = null;
    BufferedReader in = null;

    void start() {

        String hostName = "127.0.0.1";
        int portNumber = 1234;

        Socket echoSocket = null;
        try {
            echoSocket = new Socket(hostName, portNumber);
        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }

        BufferedReader stdIn = null;
        try {
            out = new PrintWriter(echoSocket.getOutputStream(), true);
            in = new BufferedReader(
                    new InputStreamReader(echoSocket.getInputStream()));
        } catch (Exception e) {
            System.err.println(e.toString());
            System.exit(1);
        }

        ClientThread clientThread = new ClientThread(in);
        clientThread.start();
    } // end of start

    void sendCmd(String cmd){
        out.println(cmd);
    }
}
```

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Network Client: send "evolve" cmd

10-SendEvolveCmd

Nel controller:

```
@FXML
protected void onEvolveButtonClick() {
    // was: String msg = new Date().toString();
    String msg = "g"; // as per AutomatonFromNetwork.
    this.virtualServer.sendCmd(msg);
    this.updateView();
}

@FXML
protected void onPayButtonClick() {
    // was: String msg = new Date().toString();
    String msg = "p"; // as per AutomatonFromNetwork.
    this.virtualServer.sendCmd(msg);
    this.updateView();
}
```

Fxml:

```
<Button text="Evolve!"
onAction="#onEvolveButtonClick"/>
<Button text="Pay!"
onAction="#onPayButtonClick"/>
```

Run...

Network Server: process and answer

11-ProcessEvolveCmdOnServer

Nel server:

```
    String s = "";
    try {
        while ((s = in.readLine()) != null) {
            System.out.println(s);
            //was: out.println(s.toUpperCase());
            out.println(processCmd(s)); // write back to client.
        }
        System.out.println("done");
    } catch (IOException e) {
        e.printStackTrace();
    }
    // we should close..
}

    static Automaton model = new Automaton();

    // from: static Boolean readLoop(... ) on Automamton code.
    static String processCmd(String s){

        String stateString;
        Boolean goOn = false;
        s = s.toUpperCase();
        System.out.println(s);
        if (s.equals("G")){
            goOn = model.evolve();
        }else if (s.equals("P")){
            model.setPaid();
        }else{
            DinnerPhase ph = DinnerPhase.fromString(s);
            goOn = model.evolveTo(ph);
        }

        stateString = model.getState().toString();
        return stateString;
    }
```

Run...

All code:

```
public class ServerMain {
    static int portNumber = 1234;

    public static void main(String[] args) {

        System.out.println("Server Started!");

        ServerSocket serverSocket = null;
        try {
            serverSocket = new ServerSocket(portNumber);
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Listening on port " + portNumber);

        Socket clientSocket = null;
        try {
            clientSocket = serverSocket.accept();
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Accepted");

        BufferedReader in = null;
        PrintWriter out = null;
        try {
            in = new BufferedReader(
                    new InputStreamReader(clientSocket.getInputStream()));
            out = new PrintWriter(clientSocket.getOutputStream(), true);

        } catch (IOException e) {
            e.printStackTrace();
        }

        String s = "";
        try {
            while ((s = in.readLine()) != null) {
                System.out.println(s);
                //was: out.println(s.toUpperCase());
                out.println(processCmd(s));

            }
            System.out.println("done");
        } catch (IOException e) {
            e.printStackTrace();
        }

        // we should close..
    }


        static Automaton model = new Automaton();

        // from: static Boolean readLoop(BufferedReader in, PrintWriter out ) on Automamton code.
        static String processCmd(String s){

            String stateString;
            Boolean goOn = false;

            s = s.toUpperCase();
            System.out.println(s);
            if (s.equals("G")){
                goOn = model.evolve();
            }else if (s.equals("P")){
                model.setPaid();
            }else{
                DinnerPhase ph = DinnerPhase.fromString(s);
                goOn = model.evolveTo(ph);
            }

            stateString = model.getState().toString();
            return stateString;
        }

}
```

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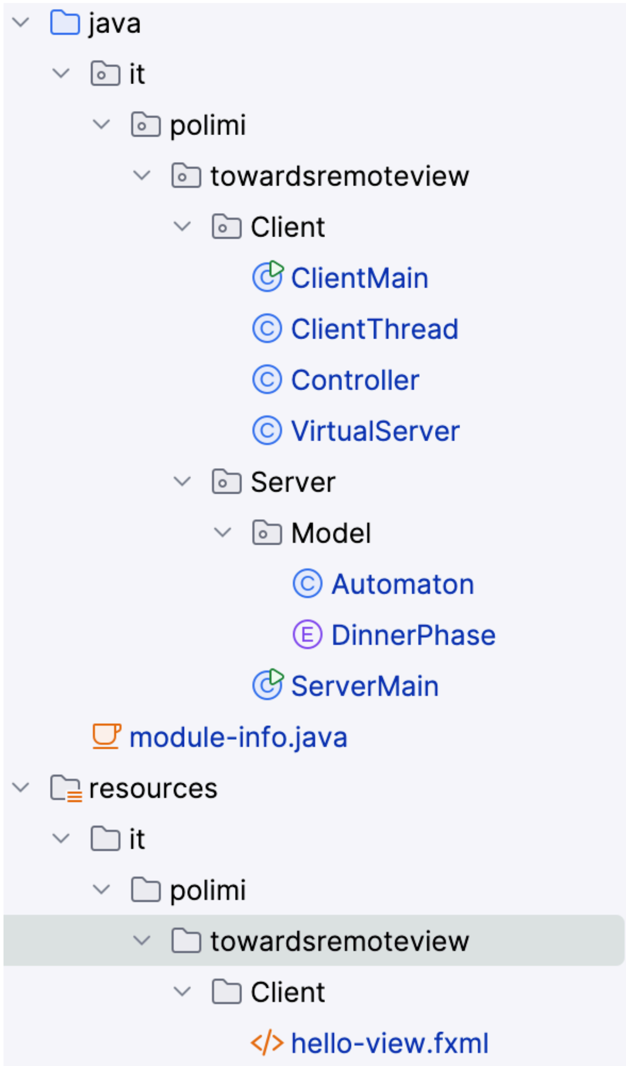
Just a bit of refactor..

12-Refactored

Separiamo in due package:



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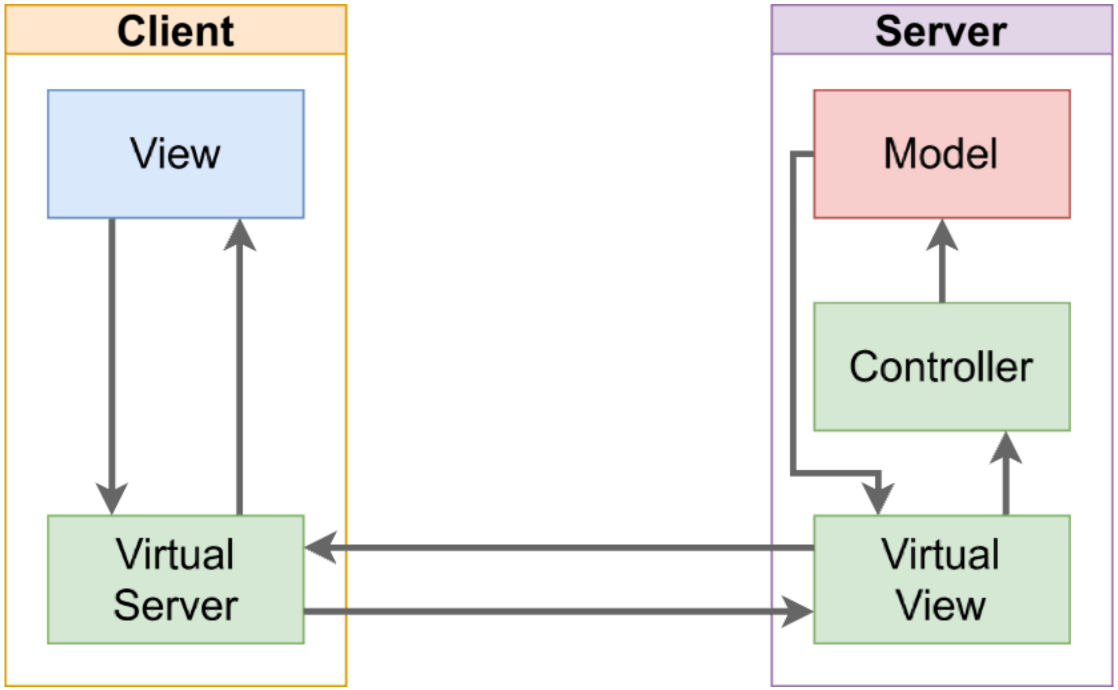
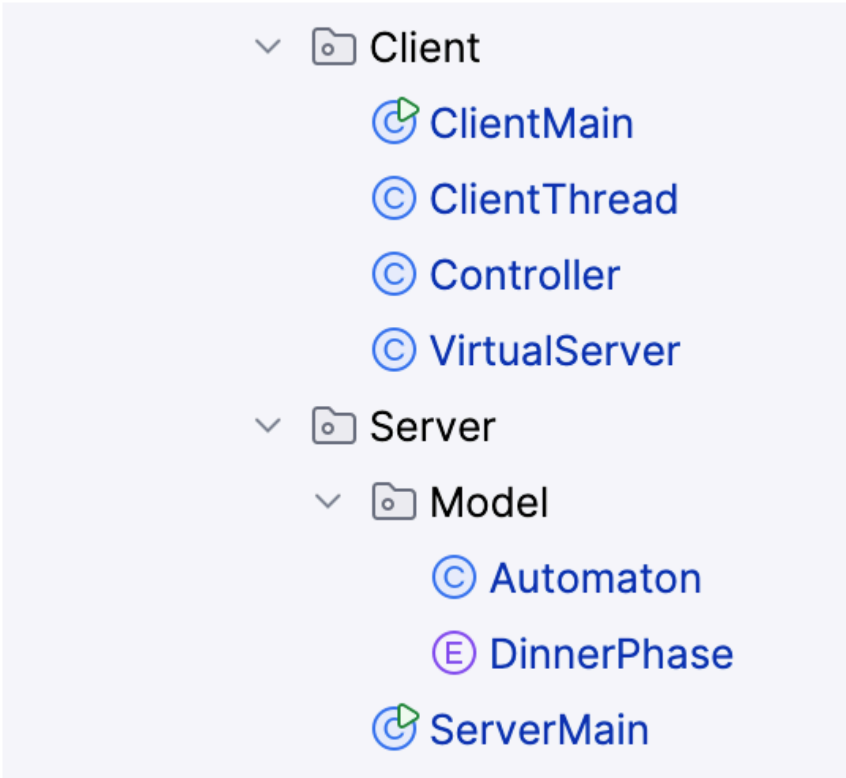
MILANO 1863

Just a moment..

Si era detto:





...

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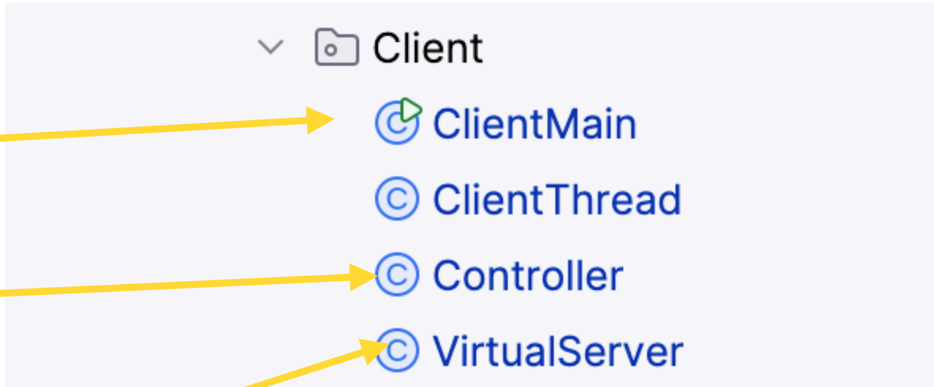
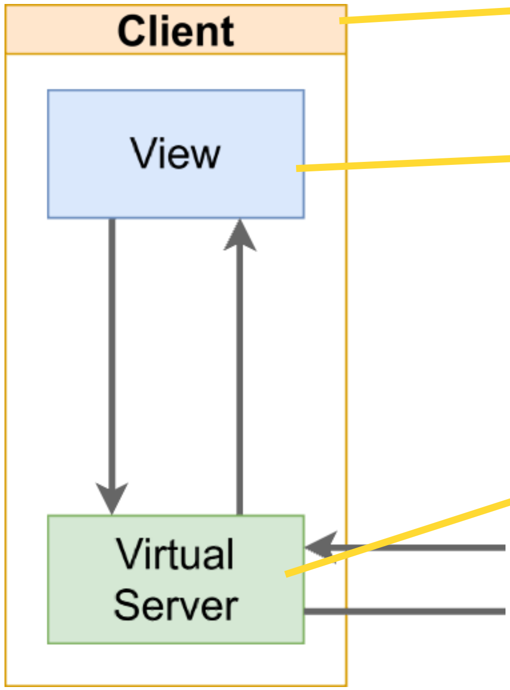
CLient ok...





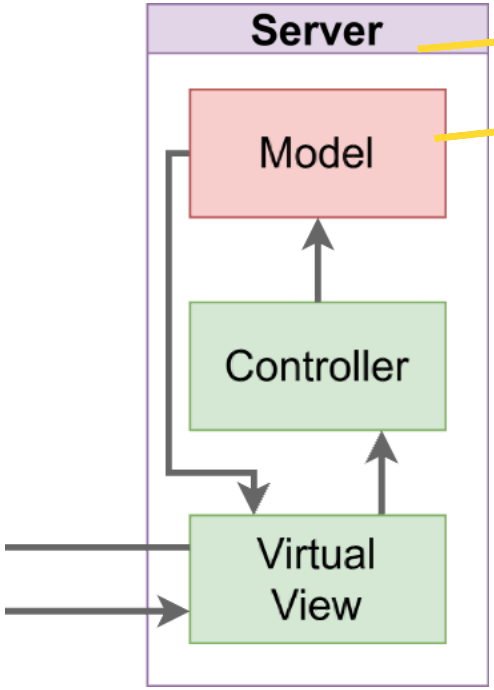
- "Controller" e' il componente JavaFX

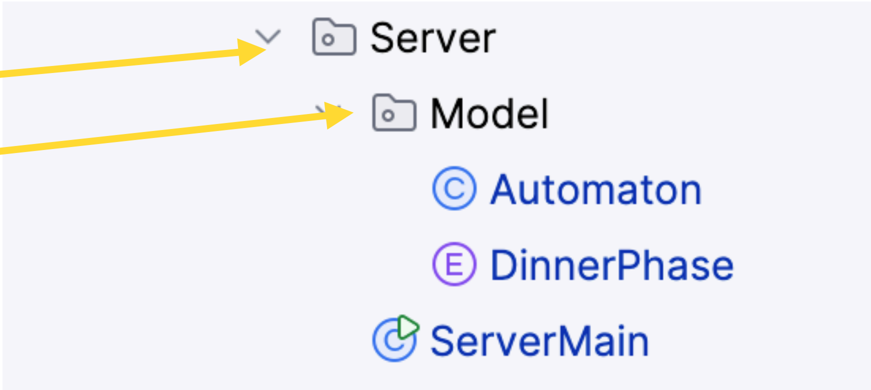
Che visualizza la UI e manda user cmds

- Il file Controller contiene implicitamente

la View

Just a moment..



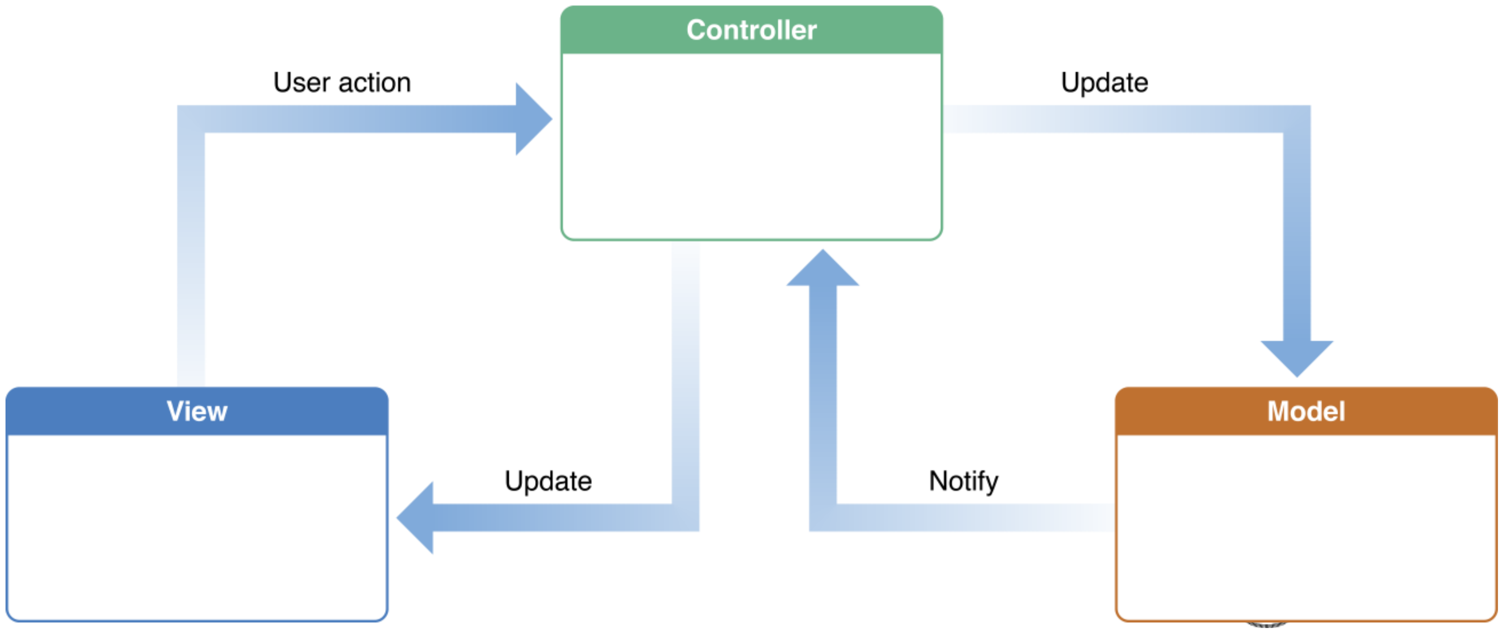


Non abbiamo separato opportunamente

Il Server...

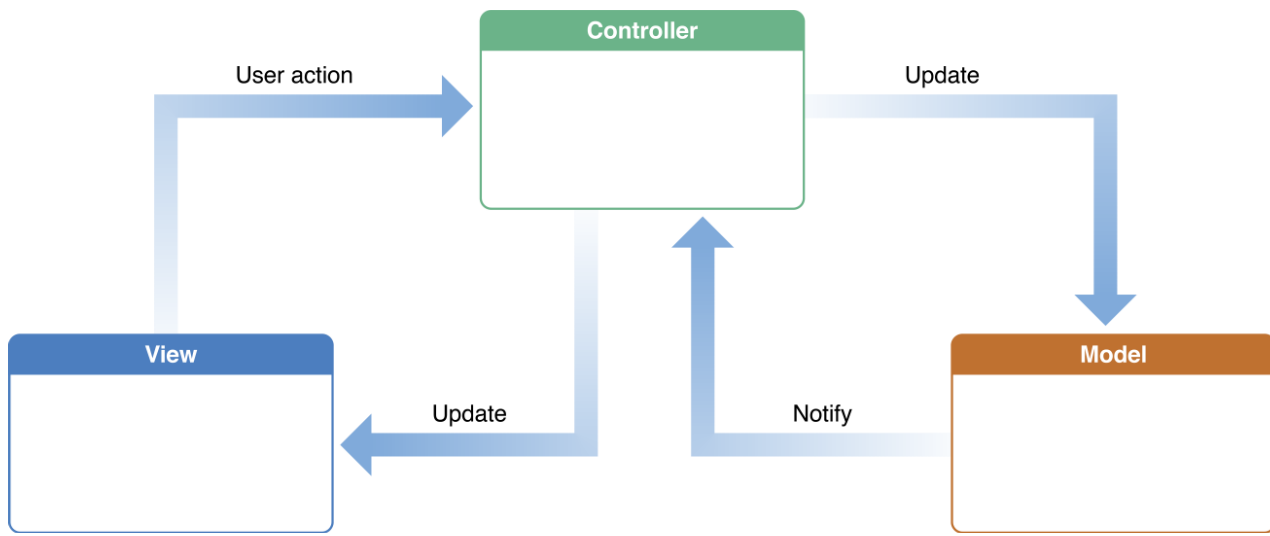
Server 1' variante

- instanzia:
- View
- Model
- Controller (a cui passa rif. a V e M)



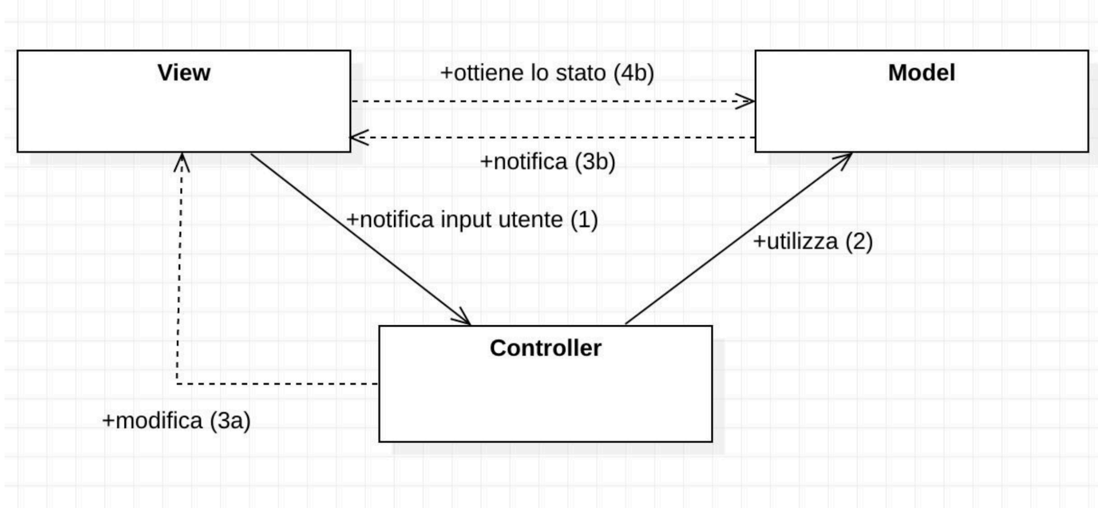
Server 1' variante: il Controller

- business logic
- Riceve input dalla View (qui dalla VV, i.e. rete, che è "virtuale")
- Modifica Modello
- É notificato dal modello (listeners) e dalla view



Server 2' variante

- instanzia:
- Model
- Controller (riceve rif. a M)
- View (riceve rif. a C)



- Listener propagano -> View (virtual) -> rete -> update (sul client) *

* slide 4

Nota: non avendo piu eventi UI,
facciamo chiamata diretta V->C
(andrebbe usati listener su UI, javaFX: public class
Controller implements ActionListener { ...)

Server 2' variante: steps

- Codice di rete nella VV
- Instanziamo VV nella App
- Nel costruttore C passiamo rif. al model (invocazione diretta)
- Nel costruttore VV passiamo rif. al controller
- **La VV non processa piu il cmd!** *(Useremo listeners)*

..

Server 2' codice VV


```
public class ServerMain {
    static int portNumber = 1234;
    static Automaton model = new Automaton();

    public static void main(String[] args) {

        System.out.println("Server Started!");
        ServerSocket serverSocket = null;
        try {
            serverSocket = new ServerSocket(portNumber);
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Listening on port " + portNumber);
        Socket clientSocket = null;
        try {
            clientSocket = serverSocket.accept();
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Accepted");

        Controller controller = new Controller(model);
        VirtualView virtualView = new VirtualView(clientSocket, controller);
        virtualView.networkEventLoop();
        // we should close..
    }
}
```

Server 2' codice Controller


```java
public class Controller {
    Automaton model = null;

    public Controller(Automaton model) {
        this.model = model;
    }

    String processCmd(String s){ // Business logic
        String stateString;
        Boolean goOn = false;

        s = s.toUpperCase();
        System.out.println(s);
        if (s.equals("G")){
            goOn = model.evolve();
        }else if (s.equals("P")){
            model.setPaid();
        }else{
            DinnerPhase ph = DinnerPhase.fromString(s);
            goOn = model.evolveTo(ph);
        }

        stateString = model.getState().toString();
        return stateString;
    }
}
```

Server 2' codice Server

13-AddedControllerAndVVOnServer

```
public class ServerMain {
    static int portNumber = 1234;
    static Automaton model = new Automaton();

    public static void main(String[] args) {

        System.out.println("Server Started!");
        ServerSocket serverSocket = null;
        try {
            serverSocket = new ServerSocket(portNumber);
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Listening on port " + portNumber);

        Socket clientSocket = null;
        try {
            clientSocket = serverSocket.accept();
        } catch (IOException e) {
            e.printStackTrace();
        }

        System.out.println("Accepted");
        Controller controller = new Controller(model);
        VirtualView virtualView = new VirtualView(clientSocket, controller);
        virtualView.networkEventLoop();
        // we should close..
    }
}
```

Run...

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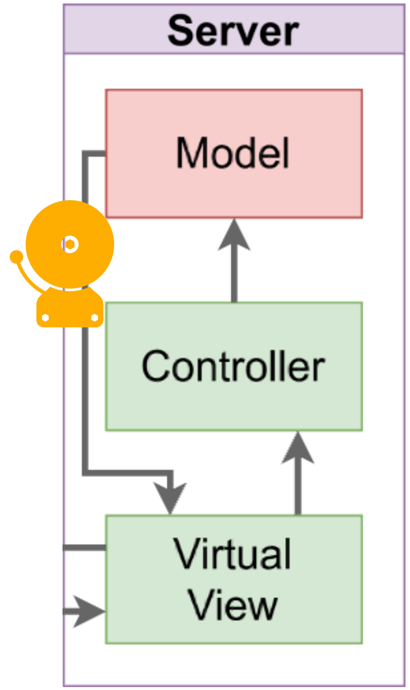
Server 2' codice Server: note

- Server riceve ma NON risponde (log ok)
- Listeners! On model
- Instanziati su VV



Listener logic...

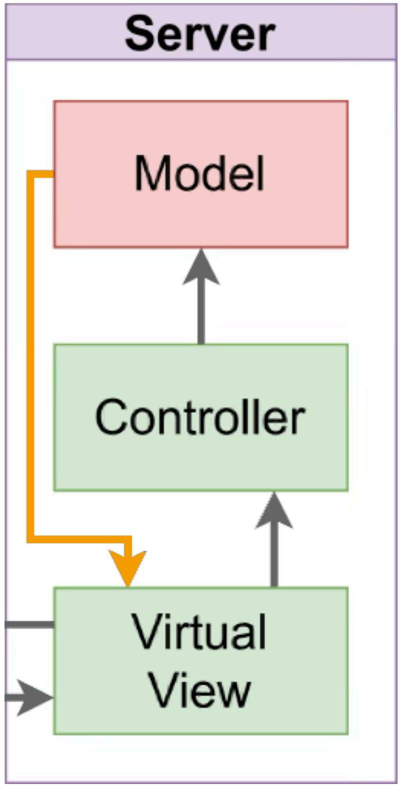
Server 2' codice Server: note

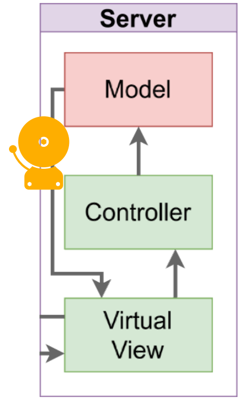
- NOTA:

Sarebbe possible anche avere:

Chiamata diretta,

ma **viola** modello ad observers/listeners





(Avremmo rif. incrociati,

M ha rif a VV, VV a M via C)

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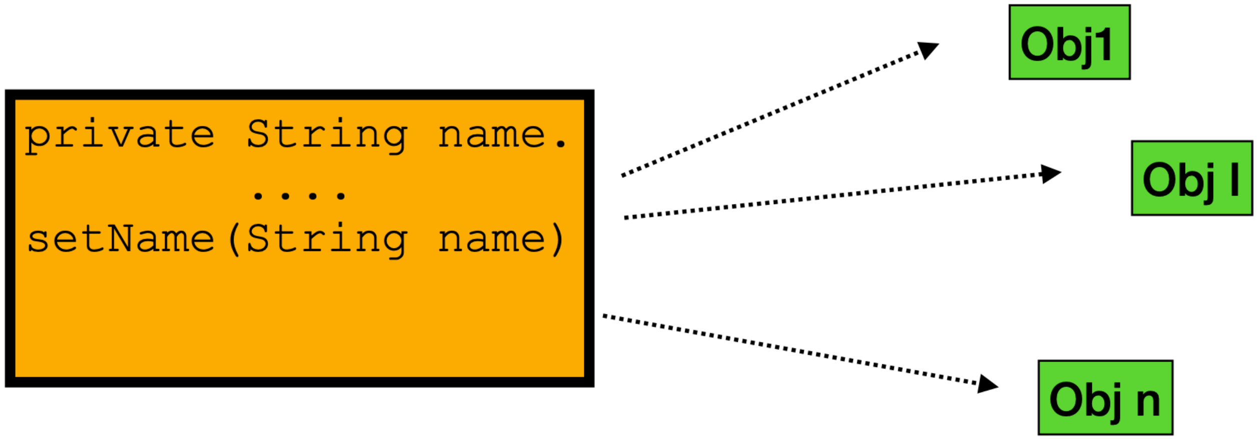
Let's listen!

Vogliamo un meccanismo che permetta a PIU oggetti di essere notificati su un cambiamento.

Perchè?
A) Per esempio vogliamo poter aggiungere **ascoltatori senza modificare il codice della sorgente** delle modifiche.
Sul "set" notifico.



B) non è necessario tenere una lista dei potenziali "interessati alle modifiche"
C) disaccoppiare le modifiche al modello dalle view che devono mostrare le modifiche.

(See also at: https://github.com/ingconti/JAVAPropertyListeners)

Let's listen!

- make attributes private *(so You are forces to pass via **Setters**)*

- Instantiate a PropertyChangeListener listener in VV

- Pass PropertyChangeListener listener to model
  (aggiungere attributo PropertyChangeListener listener; al modello)

- Set it as listener or changes in model.

Nota: by design la VV non vede il modello.. passeremo dal controller..

Model e controller

Model:

```
private PropertyChangeListener listener;

public void setListener(PropertyChangeListener listener)
{
    this.listener = listener;
}
```

Controller:

```
public void setListener(PropertyChangeListener listener)
{
    this.model.setListener(listener);
}
```

Model e controller full code

Model:

```java
public class Automaton {

    private Boolean paid = false;
    private DinnerPhase state = DinnerPhase.ENTREE;

    public DinnerPhase getState(){
        return state;
    }

    private Boolean canEvolve(){
        int currOrd = state.ordinal();
        int dessertOrd = DinnerPhase.DESSERT.ordinal();

        if (currOrd >= dessertOrd && !paid) {
            System.out.println("PAY BEFORE!!!");
            return false;
        }
        return true;
    }

    public void setPaid(){
        paid = true;
    }


    public Boolean evolve() {
        if (!canEvolve())
            return false;

        int currOrd = state.ordinal();
        int lastOrd = DinnerPhase.THE_END_OF_LUNCH.ordinal();

        if (currOrd < lastOrd) {
            state = state.next();
            return true;
        }
        return false;
    }


    public Boolean evolveTo(DinnerPhase toState){
        if (!canEvolve())
            return false;
        int toOrd = toState.ordinal();
        int currOrd = state.ordinal();

        if (toOrd>currOrd) {
            state = toState;
            return true;
        }
        return false;
    }

    private PropertyChangeListener listener;
    public void setListener(PropertyChangeListener listener) {
        this.listener = listener;
    }
}
```

Controller:

```java
public class Controller {
    Automaton model = null;

    public Controller(Automaton model) {
        this.model = model;
    }

    String processCmd(String s){ // Business logic
        String stateString;
        Boolean goOn = false;

        s = s.toUpperCase();
        System.out.println(s);
        if (s.equals("G")){
            goOn = model.evolve();
        }else if (s.equals("P")){
            model.setPaid();
        }else{
            DinnerPhase ph = DinnerPhase.fromString(s);
            goOn = model.evolveTo(ph);
        }

        stateString = model.getState().toString();
        return stateString;
    }

    public void setListener(PropertyChangeListener listener) {
        this.model.setListener(listener);
    }
}
```

Listener code 2

- *(make methods private (to model better..))*
- Instantiate a PropertyChangeListener listener in VV
- Pass PropertyChangeListener listener to model
- Set it as listener or changes in model.

Listener code


```
public class ModelListener implements PropertyChangeListener {
    @Override
    public void propertyChange(PropertyChangeEvent evt) {
        String debugStr = evt.getPropertyName()  + " from: " +
                evt.getOldValue() + " to: "  + evt.getNewValue();
        System.out.println(debugStr);
    }
}
```

VV code

14-AddingListeners

```
public class VirtualView {

    BufferedReader in = null;
    PrintWriter out = null;
    Socket clientSocket = null;
    Controller controller = null;
    ModelListener listener;

    public VirtualView(Socket clientSocket, Controller controller) {
        this.clientSocket = clientSocket;
        this.listener = new ModelListener();
        this.controller = controller;
        controller.setListener(this.listener);

        try {
            in = new BufferedReader(new InputStreamReader(clientSocket.getInputStream()));
            out = new PrintWriter(clientSocket.getOutputStream(), true);

        } catch (IOException e) {
            e.printStackTrace();
        }
    }


    void networkEventLoop(){
        String s = "";
        try {
            while ((s = in.readLine()) != null) {
                System.out.println(s);
                // no more...out.println(processCmd(s));
                String status = this.controller.processCmd(s);
                System.out.println(status); // only for debug. we do NOT send back!
            }
            System.out.println("done");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Run...

Adding Listener: RUN

14-AddingListeners

Nota: non abbiamo ancora invio indietro al client... ma sul server si vede:

```
/Users/ingconti/Library/Java/J
Server Started!
Listening on port 1234
Accepted
g
G
MAIN COURSE
g
G
SECOND COURSE
```

Ora aggiungiamo invocazione del listener sul model...

Listener - add triggering:


```
public class Automaton {

   ......
        if (currOrd < lastOrd) {
            DinnerPhase next = state.next();
            tellToListener(state, next);
            // and update:
            state = next;
            return true;
        }
        return false;
    }

    public Boolean evolveTo(DinnerPhase toState){
            ....
        if (toOrd>currOrd) {
            tellToListener(state, toState);
            // and update:
            state = toState;
            return true;
        }
        return false;
    }

    private void tellToListener(DinnerPhase from, DinnerPhase to ){
        PropertyChangeEvent evt = new PropertyChangeEvent( this, "PHASE_CHANGED", from, to);
            listener.propertyChange(evt);
    }

    private PropertyChangeListener listener;
    public void setListener(PropertyChangeListener listener) {
        this.listener = listener;
    }
}
```

Listener - add triggering: full code

15-TriggeringListeners

```
public class Automaton {

    private Boolean paid = false;
    private DinnerPhase state = DinnerPhase.ENTREE;

    public DinnerPhase getState(){
        return state;
    }

    private Boolean canEvolve(){
        int currOrd = state.ordinal();
        int dessertOrd = DinnerPhase.DESSERT.ordinal();

        if (currOrd >= dessertOrd && !paid) {
            System.out.println("PAY BEFORE!!!");
            return false;
        }
        return true;
    }

    public void setPaid(){
        paid = true;
    }


    public Boolean evolve() {
        if (!canEvolve())
            return false;

        int currOrd = state.ordinal();
        int lastOrd = DinnerPhase.THE_END_OF_LUNCH.ordinal();

        if (currOrd < lastOrd) {
            DinnerPhase next = state.next();
            tellToListener(state, next);
            // and update:
            state = next;
            return true;
        }
        return false;
    }


    public Boolean evolveTo(DinnerPhase toState){
        if (!canEvolve())
            return false;

        int toOrd = toState.ordinal();
        int currOrd = state.ordinal();

        if (toOrd>currOrd) {
            tellToListener(state, toState);
            // and update:
            state = toState;
            return true;
        }
        return false;
    }


    private void tellToListener(DinnerPhase from, DinnerPhase to ){
        PropertyChangeEvent evt = new PropertyChangeEvent(this,"PHASE_CHANGED", from, to);
            listener.propertyChange(evt);
    }

    private PropertyChangeListener listener;

    public void setListener(PropertyChangeListener listener) {
        this.listener = listener;
    }
}
```

Run...

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Adding Listener: RUN && listeners

15-TriggeringListeners

Nota: non abbiamo ancora invio indietro al client...

sul server si vede:

```
Server Started!
Listening on port 1234
Accepted
g
G
PHASE_CHANGEDENTREEMAIN COURSE
```

Ora a bit of functional to send back...

Adding Listener: RUN && listeners

15-TriggeringListeners

Nota: non abbiamo ancora invio indietro al client...

sul server si vede:

```
Server Started!
Listening on port 1234
Accepted
g
G
PHASE_CHANGEDENTREEMAIN COURSE
```

Ora a bit of functional to send back...

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Sending Back.. functional

Interfaccia:

```
public interface CallBack {
    void gotEvent(PropertyChangeEvent evt);
}
```

Sul Listener:

```
public class ModelListener implements PropertyChangeListener {
    @Override
    public void propertyChange(PropertyChangeEvent evt) {
        /*
        String debugStr = evt.getPropertyName()  + " from: " +
                evt.getOldValue() + " to: "  + evt.getNewValue();
        System.out.println(debugStr);*/
        this.callBack.gotEvent(evt);
    }

    private CallBack callBack;
    public ModelListener(CallBack callBack) {
        this.callBack = callBack;
    }
}
```

Sending Back.. functional

Interfaccia:

```
public interface CallBack {
    void gotEvent(PropertyChangeEvent evt);
}
```

Sul Listener:

```
public class ModelListener implements PropertyChangeListener {
    @Override
    public void propertyChange(PropertyChangeEvent evt) {
        /*
        String debugStr = evt.getPropertyName()  + " from: " +
                evt.getOldValue() + " to: "  + evt.getNewValue();
        System.out.println(debugStr);*/
        this.callBack.gotEvent(evt);
    }

    private CallBack callBack;
    public ModelListener(CallBack callBack) {
        this.callBack = callBack;
    }
}
```

Sending Back.. VV

Added:

```
public VirtualView(Socket clientSocket, Controller controller) {
    this.clientSocket = clientSocket;

    CallBack callBack = (PropertyChangeEvent evt) -> {
        String debugStr = evt.getPropertyName()  + " from: " +
                evt.getOldValue() + " to: "  + evt.getNewValue();
        System.out.println(debugStr);
    };

    this.listener = new ModelListener(callBack);
...
```

Sending Back.. VV full code.

16-sendingBack

```
public class VirtualView {

    BufferedReader in = null;
    PrintWriter out = null;
    Socket clientSocket = null;
    Controller controller = null;
    ModelListener listener;

    public VirtualView(Socket clientSocket, Controller controller) {
        this.clientSocket = clientSocket;

        CallBack callBack = (PropertyChangeEvent evt) -> {
            String debugStr = evt.getPropertyName()  + " from: " + evt.getOldValue() + " to: "  + evt.getNewValue();
            System.out.println(debugStr);
        };

        this.listener = new ModelListener(callBack);
        this.controller = controller;
        controller.setListener(this.listener);

        try {
            in = new BufferedReader(new InputStreamReader(clientSocket.getInputStream()));
            out = new PrintWriter(clientSocket.getOutputStream(), true);

        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    void networkEventLoop(){
        String s = "";
        try {
            while ((s = in.readLine()) != null) {
                System.out.println(s);
                // no more...out.println(processCmd(s));
                String status = this.controller.processCmd(s);
                System.out.println(status); // only for debug. we do NOT send back!
            }
            System.out.println("done");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

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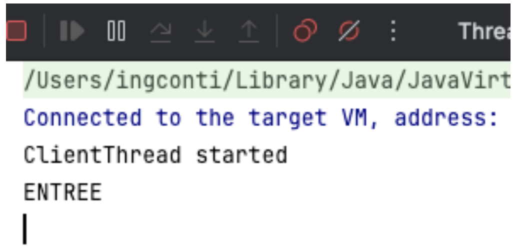
Sending Back.. network

17-sendingBack-nertwork

```java
CallBack callBack = (PropertyChangeEvent evt) -> {
    /*
    String debugStr = evt.getPropertyName()  + " from: " +
            evt.getOldValue() + " to: "  + evt.getNewValue();
    System.out.println(debugStr);
    */
    DinnerPhase state = this.controller.model.getState();
    String answer = state.toString();
    out.println(answer);
};
```

In client:



Now is time to update JavaFx UI

Update JavaFx UI

Dovremmo usare listeners sul modello, oppure (per semplicità..) avere rif. alla V (lo stage / al ctl. ..) dentro il thread.. (un po' sporco..)

Per **semplicità** / modularità:

Altra callBack:

```
public interface UpdateUICallBack {
    void process(String cmd);
}
```

Nel controller un po' di modifiche:

Update JavaFx UI: controller


```
public void setStage(Stage stage, VirtualServer virtualServer) {
    this.stage = stage;
    this.virtualServer = virtualServer;

    // add call back:
    this.updateUICallBack = new UpdateUICallBack() {
        @Override
        public void process(String cmd) {
            System.out.println("Call back " + cmd);
        }
    };

    virtualServer.setUICallBack(this.updateUICallBack);
}
```

ALL CODE:

```
public class Controller {
    @FXML
    private Label welcomeText;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

/*
    void updateView(String msg){
        this.welcomeText.setText(msg);
    }*/


    VirtualServer virtualServer;
    private Stage stage;

    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer
virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;

        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String cmd) {
                System.out.println("Call back " + cmd);
            }
        };

        virtualServer.setUICallBack(this.updateUICallBack);
    }

}
```

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Update JavaFx UI: ClientThread


```
// add call back:
private UpdateUICallBack updateUICallBack;

void setUICallBack(
    UpdateUICallBack updateUICallBack){
    this.updateUICallBack = updateUICallBack;
}
```

ALL CODE:

```
public class ClientThread extends Thread {
    private BufferedReader reader;


    public ClientThread(BufferedReader reader) {
        this.reader = reader;
    }

    public void run() {
        System.out.println("ClientThread started");

        while (true) {
            try {
                String answer = this.reader.readLine();
                System.out.println(answer);
                this.updateUICallBack.process(answer);
            } catch (IOException e) {
                e.printStackTrace();
            }
        }
    }


    // add call back:
    private UpdateUICallBack updateUICallBack;

    void setUICallBack(UpdateUICallBack updateUICallBack){
        this.updateUICallBack = updateUICallBack;
    }
}
```

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Update JavaFx UI: VirtualServer


```java
  //was: ClientThread clientThread = new ClientThread(in);
    this.clientThread = new ClientThread(in);
    //added:
    clientThread.setUICallBack(this.updateUICallBack);
    clientThread.start();
} // end of start

public void sendCmd(String cmd){
    out.println(cmd);
}


ClientThread clientThread;
private UpdateUICallBack updateUICallBack;

void setUICallBack(UpdateUICallBack updateUICallBack){
    this.updateUICallBack = updateUICallBack;
}
```

ALL CODE:

```java
public class VirtualServer {

    PrintWriter out = null;
    BufferedReader in = null;

    public void start() {

        String hostName = "127.0.0.1";
        int portNumber = 1234;

        Socket echoSocket = null;
        try {
            echoSocket = new Socket(hostName, portNumber);
        } catch (IOException e) {
            System.err.println(e.toString() + " " + hostName);
            System.exit(1);
        }

        BufferedReader stdIn = null;
        try {
            out = new PrintWriter(echoSocket.getOutputStream(), true);
            in = new BufferedReader(
                    new InputStreamReader(echoSocket.getInputStream()));
        } catch (Exception e) {
            System.err.println(e.toString());
            System.exit(1);
        }

        //was: ClientThread clientThread = new ClientThread(in);
        this.clientThread = new ClientThread(in);
        //added:
        clientThread.setUICallBack(this.updateUICallBack);
        clientThread.start();
    } // end of start

    public void sendCmd(String cmd){
        out.println(cmd);
    }


    ClientThread clientThread;
    private UpdateUICallBack updateUICallBack;

    void setUICallBack(UpdateUICallBack updateUICallBack){
        this.updateUICallBack = updateUICallBack;
    }
}
```

Update JavaFx UI: run!

18-updateStage

```
/Users/ingconti/Library/Java/JavaVirtualMachines/openj
ClientThread started
ENTREE
Call back ENTREE
MAIN COURSE
Call back MAIN COURSE
SECOND COURSE
Call back SECOND COURSE
```

(Too many debug infos....)

Update JavaFx UI: GUI

Modifichiamo callBack:

```
// add call back:
this.updateUICallBack = new UpdateUICallBack() {
    @Override
    public void process(String answer) {
        System.out.println("Call back " + answer);
        updateView(answer);
    }
};
```

Aggiungiamo:

```
void updateView(String msg){
    this.welcomeText.setText(msg);
}
```

RUN...

ALL CODE:

```
public class Controller {
    @FXML
    private Label welcomeText;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    void updateView(String msg){
        this.welcomeText.setText(msg);
    }


    VirtualServer virtualServer;
    private Stage stage;

    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;

        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String answer) {
                System.out.println("Call back " + answer);
                updateView(answer);
            }
        };

        virtualServer.setUICallBack(this.updateUICallBack);
    }

}
```

Update JavaFx UI: GUI **CRASH**

`FIX: RunLater!`

Update JavaFx UI: GUI

19-updateStage-GUI-fixed_crash

FIX:

```java
private void safeUpdateView(String msg){
    Platform.runLater(new Runnable() {
        @Override
        public void run() {
            updateView(msg);
        }
    });

}

private void updateView(String msg){
    this.welcomeText.setText(msg);
}
```

...

```java
public void process(String answer) {
    System.out.println("Call back " + answer);
    safeUpdateView(answer);
}
```

RUN.. ok

(Non gestiamo x per ora errore da automa)

ALL CODE:

```java
public class Controller {
    @FXML
    private Label welcomeText;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }


    private void safeUpdateView(String msg){
        Platform.runLater(new Runnable() {
            @Override
            public void run() {
                updateView(msg);
            }
        });

    }

    private void updateView(String msg){
        this.welcomeText.setText(msg);
    }


    VirtualServer virtualServer;
    private Stage stage;

    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;

        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String answer) {
                System.out.println("Call back " + answer);
                safeUpdateView(answer);
            }
        };

        virtualServer.setUICallBack(this.updateUICallBack);
    }

}
```

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Update JavaFx UI : Drawing in graphics

FXML, aggiungiamo una hBox:

```
<HBox fx:id="hbox" >

</HBox>
```

Controller, aggiungiamo una hBox:

```
@FXML
private Label welcomeText;

@FXML
private HBox hbox;
```

Update JavaFx UI : Drawing in graphics (2)

Controller:

```
private void updateView(String msg){
    this.welcomeText.setText(msg);
    this.addCircle(msg);
}
```

+ altri metodi..

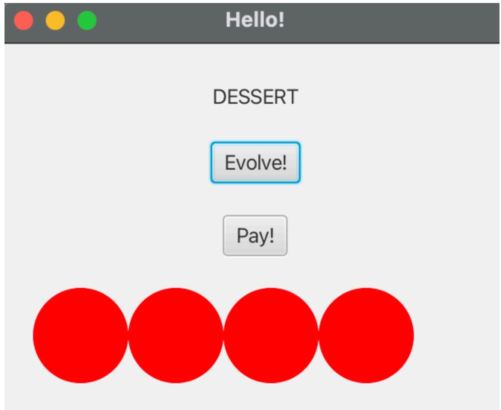
```
private void addCircle(String answer){
    double centerX = xForAnswer(answer);
    double centerY = 100;
    double radius = 30;

    Circle c1 = new Circle(centerX, centerY, radius, Color.RED);
    this.hbox.getChildren().add(c1);
}
```

Faremo apparire dei cerchi, uno x fase....

Update JavaFx UI : Drawing in graphics full code

```java
public class Controller {
    @FXML
    private Label welcomeText;
    @FXML
    private HBox hbox;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

   private int xForAnswer(String answer) {     // todo: You will process answer..
        switch (answer) {
            case "ENTREE":return 30;
            case "MAIN COURSE":return 60;
            case "SECOND COURSE":return 90;
            case "DESSERT":return 120;
            case "END OF YOUR LUNCH!":return 30;
        }
        return 0;
    }

    private void addCircle(String answer){
        double centerX = xForAnswer(answer), centerY = 100, radius = 30;

        Circle c1 = new Circle(centerX, centerY, radius, Color.RED);
        this.hbox.getChildren().add(c1);
    }

    private void safeUpdateView(String msg){
        Platform.runLater(new Runnable() {
            @Override
            public void run() {
                updateView(msg);
            }
        });

    }

    private void updateView(String msg){
        this.welcomeText.setText(msg);
        this.addCircle(msg);
    }


    VirtualServer virtualServer;
    private Stage stage;
    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;

        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String answer) {
                System.out.println("Call back " + answer);
                safeUpdateView(answer);
            }
        };

        virtualServer.setUICallBack(this.updateUICallBack);
    }
}
```

Run...



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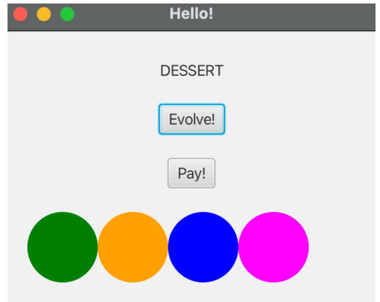
Update JavaFx UI : Drawing.. colors..

21-updateStage-GUI-ColoredCircles

```java
public class Controller {
    @FXML
    private Label welcomeText;
    @FXML
    private HBox hbox;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    // todo: You will process answer..
    private ColorsAndCoord ColorsAndCoordForAnswer(String answer) {
        switch (answer) {
            case "ENTREE":return new ColorsAndCoord(30, Color.GREEN);
            case "MAIN COURSE":return new ColorsAndCoord(60, Color.ORANGE);
            case "SECOND COURSE":return new ColorsAndCoord(90, Color.BLUE);
            case "DESSERT":return new ColorsAndCoord(120, Color.MAGENTA);
            case "END OF YOUR LUNCH!":return new ColorsAndCoord(150, Color.YELLOW);
        }
        return new ColorsAndCoord(30, Color.BLACK);
    }

    private void addCircle(String answer){
        ColorsAndCoord colorsAndCoord = ColorsAndCoordForAnswer(answer);
        double centerX = colorsAndCoord.x, centerY = 100, radius = 30;
        Circle c1 = new Circle(centerX, centerY, radius, colorsAndCoord.c);
        this.hbox.getChildren().add(c1);
    }

    private void safeUpdateView(String msg){
        Platform.runLater(new Runnable() {
            @Override
            public void run() {
                updateView(msg);
            }
        });

    }

    private void updateView(String msg){
        this.welcomeText.setText(msg);
        this.addCircle(msg);
    }

    VirtualServer virtualServer;
    private Stage stage;
    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;

        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String answer) {
                System.out.println("Call back " + answer);
                safeUpdateView(answer);
            }
        };
        virtualServer.setUICallBack(this.updateUICallBack);
    }
}
```

Run...



Update JavaFx UI: all by code...

21-updateStage-GUI-ColoredCircles

Potremmo anche non usare del tutto FXML e usare solo codice x disegnare:

Al messaggio "DESSERT" svuotiamo tutto *(Dessert appare solo dopo aver pagato.. 3 Evolve->Pay->Evolve)*

```
private void cleanUpAll() {
    Group root = new Group();
    Canvas canvas = new Canvas(300, 250);
    GraphicsContext gc = canvas.getGraphicsContext2D();
    drawShapes(gc);
    root.getChildren().add(canvas);
    stage.setScene(new Scene(root));
    stage.show();
}

private void safeCleanUpAll() {
    Platform.runLater(new Runnable() {
        @Override
        public void run() {
            cleanUpAll();
        }
    });
}

private void drawShapes(GraphicsContext gc) {
    gc.setFill(Color.GREEN);
    gc.setStroke(Color.BLUE);
    gc.setLineWidth(5);
    gc.strokeLine(40, 10, 10, 40);
    gc.fillOval(10, 60, 30, 30);
    gc.strokeOval(60, 60, 30, 30);
    gc.fillRoundRect(110, 60, 30, 30, 10, 10);
    gc.strokeRoundRect(160, 60, 30, 30, 10, 10);
    gc.fillArc(10, 110, 30, 30, 45, 240, ArcType.OPEN);
    gc.fillArc(60, 110, 30, 30, 45, 240, ArcType.CHORD);
    gc.fillArc(110, 110, 30, 30, 45, 240, ArcType.ROUND);
    gc.strokeArc(10, 160, 30, 30, 45, 240, ArcType.OPEN);
    gc.strokeArc(60, 160, 30, 30, 45, 240, ArcType.CHORD);
    gc.strokeArc(110, 160, 30, 30, 45, 240, ArcType.ROUND);
    gc.fillPolygon(new double[]{10, 40, 10, 40}, new double[]{210, 210, 240, 240}, 4);
    gc.strokePolygon(new double[]{60, 90, 60, 90},new double[]{210, 210, 240, 240}, 4);
    gc.strokePolyline(new double[]{110, 140, 110, 140},new double[]{210, 210, 240, 240}, 4);
}
```

```
// add call back:
this.updateUICallBack = new UpdateUICallBack() {
    @Override
    public void process(String answer) {
        System.out.println("Call back " + answer);
        safeUpdateView(answer);
        if (answer.equals("DESSERT")){
            safeCleanUpAll();
        }
    }
};
```

Update JavaFx UI: all by code. All Code

21-updateStage-GUI-ColoredCircles

```java
public class Controller {
    @FXML
    private Label welcomeText;
    @FXML
    private HBox hbox;

    @FXML
    protected void onEvolveButtonClick() {
        String msg = "g"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

    @FXML
    protected void onPayButtonClick() {
        String msg = "p"; // as per AutomatonFromNetwork.
        this.virtualServer.sendCmd(msg);
    }

   private ColorsAndCoord ColorsAndCoordForAnswer(String answer) {
        switch (answer) {
            case "ENTREE":return new ColorsAndCoord(30, Color.GREEN);
            case "MAIN COURSE":return new ColorsAndCoord(60, Color.ORANGE);
            case "SECOND COURSE":return new ColorsAndCoord(90, Color.BLUE);
            case "DESSERT":return new ColorsAndCoord(120, Color.MAGENTA);
            case "END OF YOUR LUNCH!":return new ColorsAndCoord(150, Color.YELLOW);
        }
        return new ColorsAndCoord(30, Color.BLACK);
    }

    private void addCircle(String answer){
        ColorsAndCoord colorsAndCoord = ColorsAndCoordForAnswer(answer);
        double centerX = colorsAndCoord.x, centerY = 100, radius = 30;
        Circle c1 = new Circle(centerX, centerY, radius, colorsAndCoord.c);
        this.hbox.getChildren().add(c1);
    }

    private void safeUpdateView(String msg){
        Platform.runLater(new Runnable() {
            @Override
            public void run() {
                updateView(msg);
            }
        });
    }

    private void updateView(String msg){
        this.welcomeText.setText(msg);
        this.addCircle(msg);
    }

    VirtualServer virtualServer;
    private Stage stage;
    UpdateUICallBack updateUICallBack;

    public void setStage(Stage stage, VirtualServer virtualServer) {
        this.stage = stage;
        this.virtualServer = virtualServer;
        // add call back:
        this.updateUICallBack = new UpdateUICallBack() {
            @Override
            public void process(String answer) {
                System.out.println("Call back " + answer);
                safeUpdateView(answer);
                if (answer.equals("DESSERT")){
                    safeCleanUpAll();
                }
            }
        };
        virtualServer.setUICallBack(this.updateUICallBack);
    }

    private void cleanUpAll() {
        Group root = new Group();
        Canvas canvas = new Canvas(300, 250);
        GraphicsContext gc = canvas.getGraphicsContext2D();
        drawShapes(gc);
        root.getChildren().add(canvas);
        stage.setScene(new Scene(root));
        stage.show();
    }

    private void safeCleanUpAll() {
        Platform.runLater(new Runnable() {
            @Override
            public void run() {
                cleanUpAll();
            }
        });
    }

    private void drawShapes(GraphicsContext gc) {
        gc.setFill(Color.GREEN);
        gc.setStroke(Color.BLUE);
        gc.setLineWidth(5);
        gc.strokeLine(40, 10, 10, 40);
        gc.fillOval(10, 60, 30, 30);
        gc.strokeOval(60, 60, 30, 30);
        gc.fillRoundRect(110, 60, 30, 30, 10, 10);
        gc.strokeRoundRect(160, 60, 30, 30, 10, 10);
        gc.fillArc(10, 110, 30, 30, 45, 240, ArcType.OPEN);
        gc.fillArc(60, 110, 30, 30, 45, 240, ArcType.CHORD);
        gc.fillArc(110, 110, 30, 30, 45, 240, ArcType.ROUND);
        gc.strokeArc(10, 160, 30, 30, 45, 240, ArcType.OPEN);
        gc.strokeArc(60, 160, 30, 30, 45, 240, ArcType.CHORD);
        gc.strokeArc(110, 160, 30, 30, 45, 240, ArcType.ROUND);
        gc.fillPolygon(new double[]{10, 40, 10, 40}, new double[]{210, 210, 240, 240}, 4);
        gc.strokePolygon(new double[]{60, 90, 60, 90},new double[]{210, 210, 240, 240}, 4);
        gc.strokePolyline(new double[]{110, 140, 110, 140}, new double[]{210, 210, 240, 240}, 4);
    }
}
```

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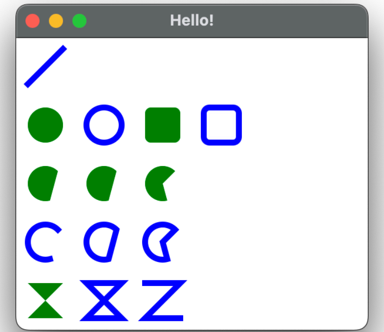
Update JavaFx UI: all by code...

Potremmo anche non usare del tutto FXML e usare solo codice x disegnare:

Al messaggio "DESSERT" svuotiamo tutto *(Dessert appare solo dopo aver pagato.. 3 Evolve->Pay->Evolve)*



End...